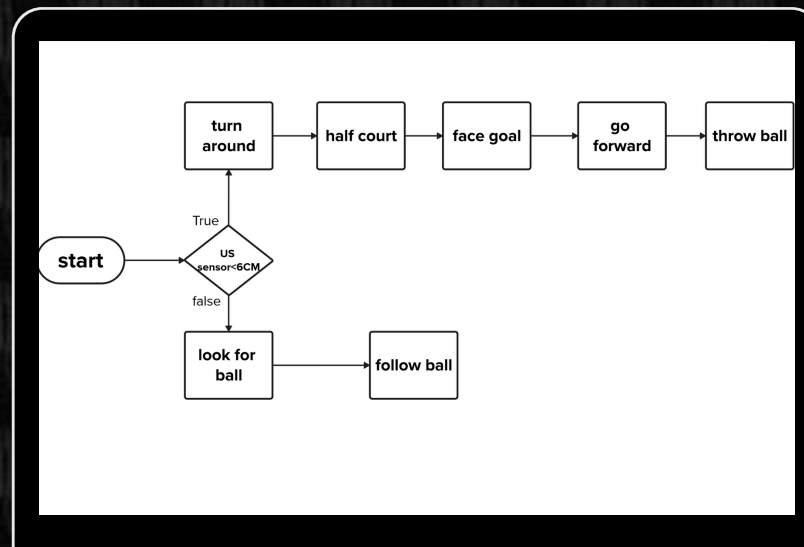


ZAPPEGONI

SOCCER ENTRY TEAM ITALY NATIONAL CHAMPIONS

SOFTWARE



To program our robot we used the iconic EV3 language, with the Lego Mindstorm Education software.

This allowed us to have full control of our hardware and in the most intuitive way

The robot knows if he has the ball or not thanks to the UltraSonic sensor in the front, if the sensor reads less than 6CM the robot has the ball in the dribbler, that starts rotating fast to keep control of it. Then the robot turns facing the longer wall and gets to half field, to then turn around facing the goal and go forward to score the goal.

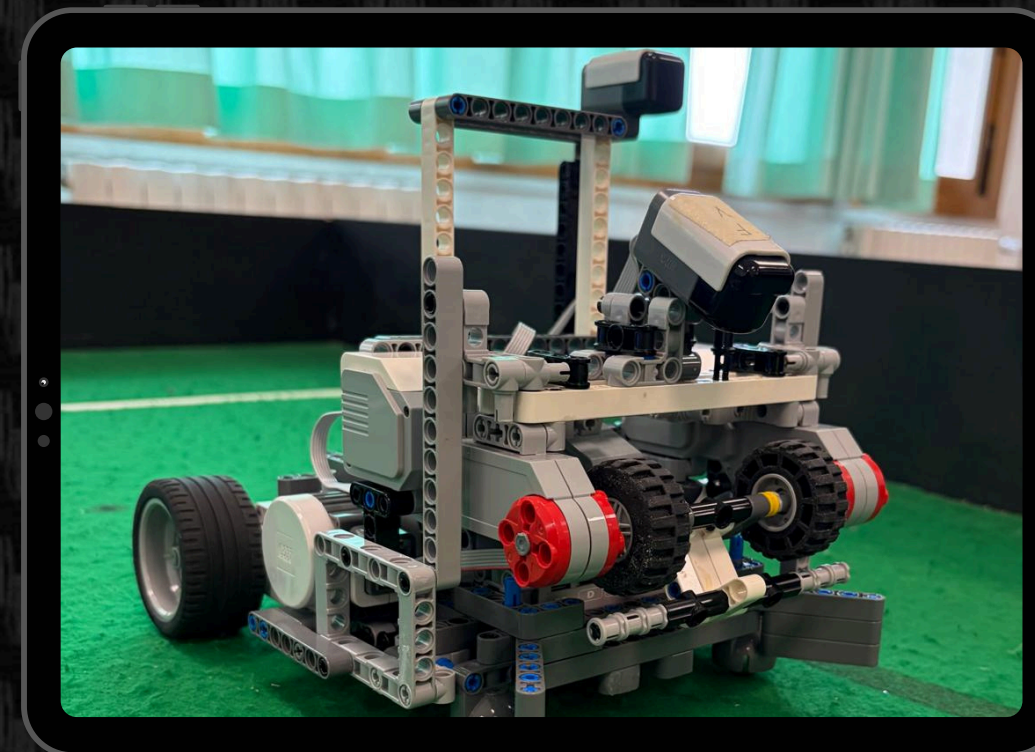
If the robot doesn't have the ball, he keeps looking for it with the InfraRed seeker and follows it, until he gets the ball.

Focal points:

- Ball recognition with InfraRed seeker
- Ball possess with Color sensor
- Compass and Camera orientation

HARDWARE

ROBOT LEGO EV3



Our robot is fully built with the Lego ev3 environment, allowing us to understand what we are doing even as beginners.

We decided to build a dribbler to have a steady control of the ball, using two motors and some wheels of different sizes to keep the ball in the center. We also implemented all of the things below:



HITECHNIC EV3

The brain of our robot, where all the sensors are connected



NXT LEGO LARGE MOTORS

Lego motors, used not only to move the robot, but also to rotate the dribbler.



HITECHNIC IR SEEKER

One of the "eyes" of our robot, used to locate the ball.



HITECHNIC COMPASS

This compass allows the robot to know where he is and where to go



COLOR SENSOR

Used to let the robot know that he got the ball in the dribbler thanks to the light reflection.



CAMERA

Pixy camera used to aim precisely at the goal, calibrated to have 0 errors

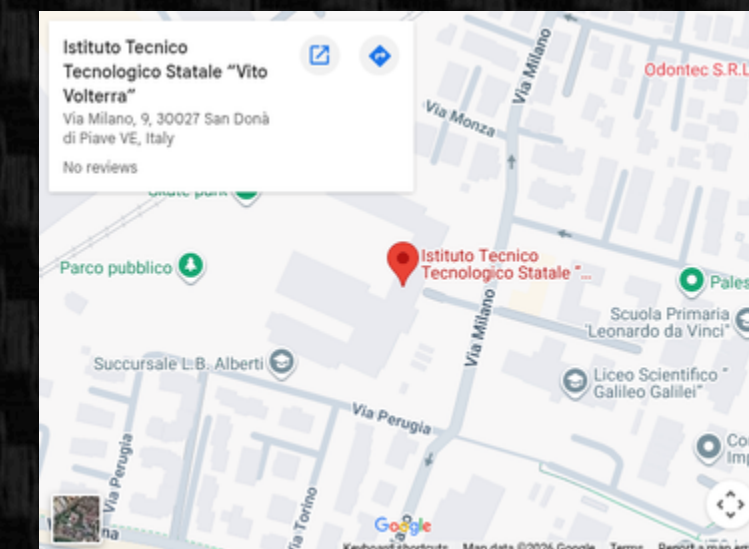
ABOUT US

We're a team of young students from San Donà Di Piave, near Venice, in Veneto.

We're two students of the Vito Volterra ITTS, from the Automation specialization.

It's our first year participating in the robocup. We started building our robot in November 2025 and since then we learned how to work in group on cooperate to be efficient.

We also won the national races in Catania, qualifying first.



**OFFICIAL
SPONSOR**



BACCICHETTO ALBERTO

Leader
Took care of the structure and the program.



FANTIN DAVIDE

Co-Leader
Took care of the sensors and motors.